

Red Stealer Lab

Analyze a suspicious executable using VirusTotal and MalwareBazaar to extract IOCs, identify C2 infrastructure, MITRE ATT&CK techniques, and privilege escalation mechanisms.

Category:

[Threat Intel](#)

Tactics:

[Execution](#)[Persistence](#)[Privilege Escalation](#)[Defense Evasion](#)[Discovery](#)[Collection](#)[Impact](#)

Tools:

[Whois](#)[VirusTotal](#)[MalwareBazaar](#)[ThreatFox](#)[ANY.RUN](#)

Scenario

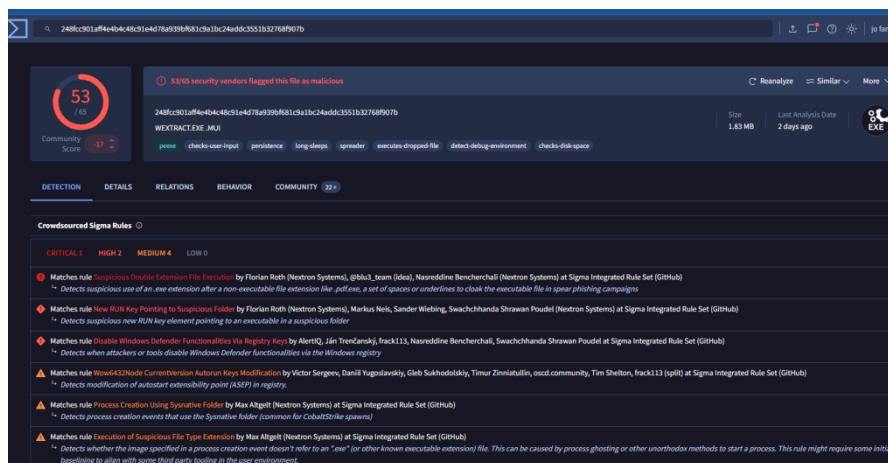
You are part of the Threat Intelligence team in the SOC (Security Operations Center). An executable file has been discovered on a colleague's computer, and it's suspected to be linked to a Command and Control (C2) server, indicating a potential malware infection.

Your task is to investigate this executable by analyzing its hash. The goal is to gather and analyze data beneficial to other SOC members, including the Incident Response team, to respond to this suspicious behavior efficiently.

File Hash (SHA-256):

248FCC901AFF4E4B4C48C91E4D78A939BF681C9A1BC24ADDC3551B32768F907B

<https://www.virustotal.com/gui/file/248fcc901aff4e4b4c48c91e4d78a939bf681c9a1bc24addc3551b32768f907b>



Q1

Solved : 7269

Categorizing malware enables a quicker and clearer understanding of its unique behaviors and attack vectors. What category has Microsoft identified for that malware in VirusTotal?

Threat categorie; Trojan

Q2

Solved : 7185

Clearly identifying the name of the malware file improves communication among the SOC team. What is the file name associated with this malware?

Note: Don't include the file extension in the name.

WEXTRACT

Q3
Solved : 7092

Knowing the exact timestamp of when the malware was first observed can help prioritize response actions. Newly detected malware may require urgent containment and eradication compared to older, well-documented threats. What is the UTC timestamp of the malware's first submission to VirusTotal?

2023-10-06 04:41

Q4
Solved : 6964

Understanding the techniques used by malware helps in strategic security planning. What is the MITRE ATT&CK technique ID for the malware's data collection from the system before exfiltration?

T1005

- **Go to the "Behavior" tab** → this shows dynamic analysis results from sandboxes
- **Scroll down to "MITRE ATT&CK Techniques"** → VT maps observed behaviors to the ATT&CK framework automatically
- **Look for the "Collection" tactic** → since the question asks about *data collection before exfiltration*, you're looking for techniques under the **Collection** tactic category

Technique ID	Name
T1005	Data from Local System
T1113	Screen Capture
T1115	Clipboard Data
T1056	Input Capture
T1119	Automated Collection

Tips

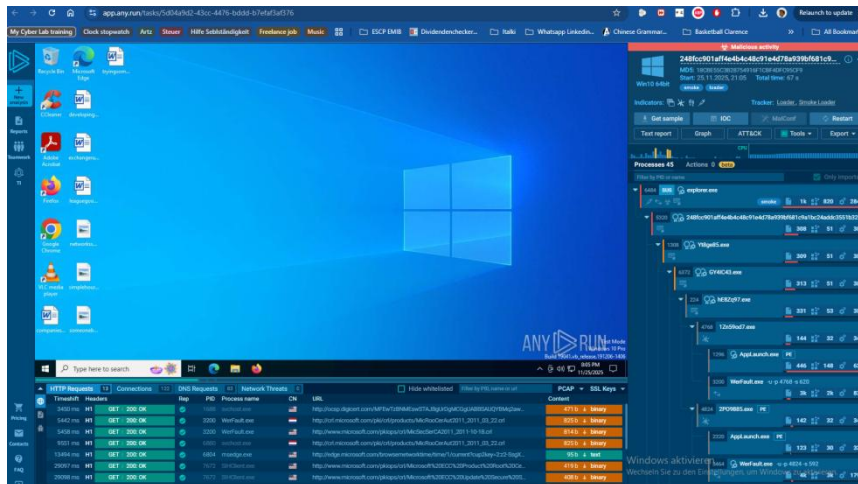
- In the MITRE ATT&CK section on VT, techniques are **color-coded by tactic** — look for the **orange/yellow "Collection"** category
- The question specifically says "*before exfiltration*" — so focus on **Collection**, not **Exfiltration** techniques

Cross-reference with **ANY.RUN** sandbox if you need more detail

<https://app.any.run/submissions>

248fcc901aff4e4b4c48c91e4d78a939bf681c9a1bc24addc3551b32768f907b

<https://app.any.run/tasks/5d04a9d2-43cc-4476-bddd-b7efaf3af376>



Q5

Solved : 6890

Following execution, which social media-related domain names did the malware resolve via DNS queries?

- **Go to the "Behavior" tab**
- **Scroll down to "Network Communication" section**
- **Look for "DNS Resolutions" or "Domains" subsection** → this lists all domains the malware queried via DNS during sandbox execution
- **Filter visually for social media domains** → look for anything resembling:

- facebook.com, twitter.com, instagram.com
- discord.com, telegram.org
- reddit.com, youtube.com

facebook.com

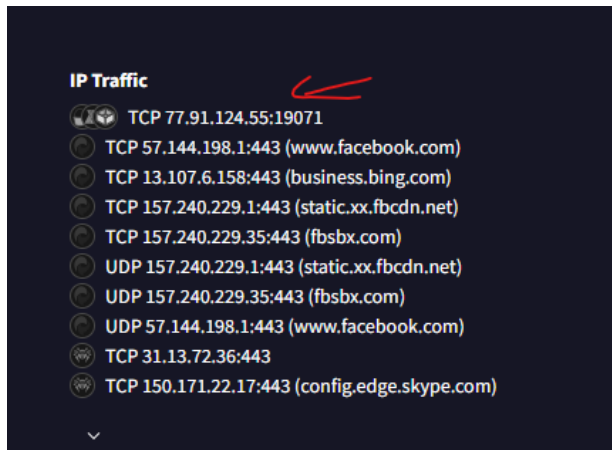
Pro Tips

- You can also check the **"Relations" tab** → shows all domains/IPs the sample communicates with, sometimes easier to browse
- In **ANY.RUN**, go to the **"Network"** tab → DNS requests are listed in real time during execution, very visual and easy to read
- In **MalwareBazaar**, DNS info is limited — VT and ANY.RUN are your best sources for this

Q6

Solved : 6836

Once the malicious IP addresses are identified, network security devices such as firewalls can be configured to block traffic to and from these addresses. Can you provide the IP address and destination port the malware communicates with?



TCP 77.91.124.55:19071

Q7

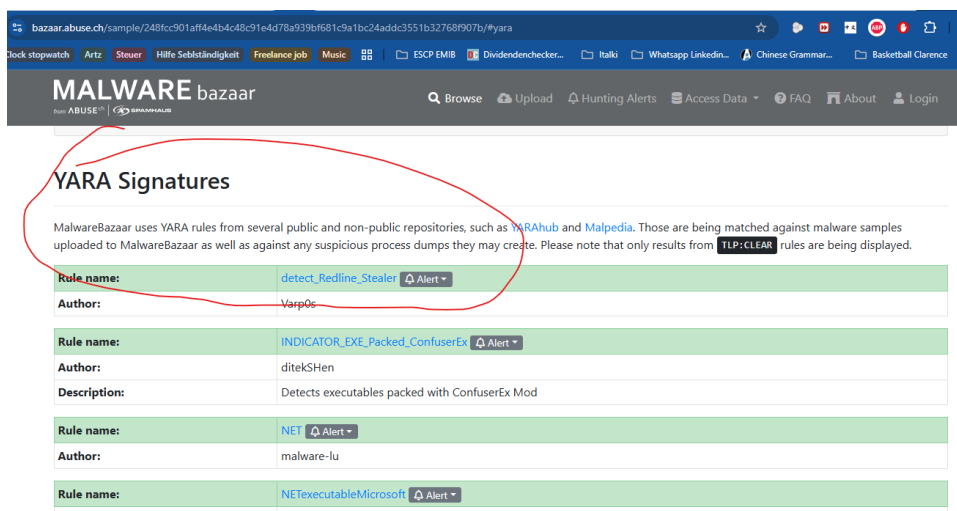
Solved : 6631

YARA rules are designed to identify specific malware patterns and behaviors. Using MalwareBazaar, what's the name of the YARA rule created by "Varp0s" that detects the identified malware?

Correct Steps in MalwareBazaar

1. Go to `bazaar.abuse.ch`
2. Search by your hash (MD5 or SHA256) directly in the search bar → not by "Varp0s"
3. Click on the result to open the sample details page
4. Scroll down to the "YARA Signatures" section
5. Look for the rule where the author is "Varp0s"

<https://bazaar.abuse.ch/sample/248fcc901aff4e4b4c48c91e4d78a939bf681c9a1bc24addc3551b32768f907b/>



[detect Redline Stealer](#)

Q8
Solved : 6458

Understanding which malware families are targeting the organization helps in strategic security planning for the future and prioritizing resources based on the threat. Can you provide the different malware alias associated with the **malicious IP address** according to **ThreatFox**?

Steps in ThreatFox

- Go to <https://threatfox.abuse.ch>
- Look for the "IOCs" menu at the top
- Click "Search IOCs" or look for a **magnifying glass icon**
- Or try the direct URL:

<https://threatfox.abuse.ch/browse.php?q=77.91.124.55>

RECORDSTEALER

Q9
Solved : 6520

By identifying the malware's imported DLLs, we can configure security tools to monitor for the loading or unusual usage of these specific DLLs. Can you provide the DLL utilized by the malware for privilege escalation?

<https://www.virustotal.com/gui/file/248fcc901aff4e4b4c48c91e4d78a939bf681c9a1bc24addc3551b32768f907b/details>

- Click the "Details" tab
- Scroll down to "Imports" section
 - This lists all **DLLs and functions** imported by the malware
- Look for DLLs associated with privilege escalation **Search .dll**



What to Look For

Focus on these suspicious DLLs commonly used for privilege escalation:

DLL	Purpose
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DLL	Purpose
advapi32.dll	Token manipulation, registry access
ntdll.dll	Low-level Windows API, process injection
shell32.dll	UAC bypass techniques
kernel32.dll	Process/memory manipulation
psapi.dll	Process enumeration

Cross-Reference with MITRE

Once you find the DLL, match it to:

- **T1055** → Process Injection → `ntdll.dll`
- **T1548.002** → UAC Bypass → `shell32.dll`
- **T1134** → Token Impersonation → `advapi32.dll`

VT **Behavior tab** → MITRE ATT&CK section → filter by "**Privilege Escalation**" tactic → the technique there will hint at which DLL

Response: ADVAPI32.dll

Red Stealer Lab — Investigation Summary [?](#)

Malware Profile

Field	Details
SHA-256	248fcc901aff4e4b4c48c91e4d78a939bf681c9a1bc24addc3551b32768f907b
File Name	WEXTRACT
Category	Trojan
First Seen	2023-10-06 04:41 UTC

IOCs Identified

Type	Value
C2 IP:Port	77.91.124.55:19071

Type	Value
DNS Query	facebook.com
Malware Alias	RECORDSTEALER

MITRE ATT&CK & Techniques

Tactic	Technique ID	Details
Collection	T1005	Data from Local System
Privilege Escalation	T1134	Token Impersonation via ADVAPI32.dll

Forensic Findings

Tool	Finding
VirusTotal	Trojan, DLL imports, MITRE mapping
MalwareBazaar	YARA rule: detect_Redline_Stealer by Varp0s
ThreatFox	Alias: RECORDSTEALER
ANY.RUN	DNS queries, network traffic

Key Takeaways

- This is a **RedLine/Record Stealer** variant
- Uses **facebook.com** DNS queries possibly for connectivity checks or C2 dead drop
- Communicates with **77.91.124.55:19071** for C2
- Escalates privileges via **ADVAPI32.dll** token manipulation
- Detected by YARA rule **detect_Redline_Stealer**